

Algebraic Topology 1

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MA727

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Abstract

TODO

Chapter 1

The fundamental group

Lecture 1: Syllabus day

Thu 03 Sep 2026 09:31

We are following Hatcher. Takashi's office is probably CDS 415, or at least in that area. Half the grade will be homework, and the other half will be a final paper/presentation. Assignments will be submitted on gradescope. It is unknown if a syllabus will be made for the course.

The basic idea of algebraic topology is to associate a space X to some algebra $H_\bullet(X)$. We want this association to be functorial, i.e. if $f : X \rightarrow Y$, then there is a map $f_* : H_\bullet(X) \rightarrow H_\bullet(Y)$, and $(g \circ f)_* = g_* \circ f_*$. There are a number of ways to do this, namely homology and cohomology (cohomology is contravariant though, meaning $(g \circ f)^* = f^* \circ g^*$).

1.1 Basic constructions

1.1.1 Paths and homotopy

I am extremely well-aquainted with homotopy, and choose not to write down much related to it. I have listed only the following useful formula for linear homotopies.

Given $x_0, x_1 \in \mathbb{R}^n$ and paths $f_0, f_1 : x_0 \rightarrow x_1$, we define

$$H(s, t) = (1 - t)f_0(s) + tf_1(s).$$

This is a homotopy $H : f_0 \Rightarrow f_1$. This construction is easily seen to hold in all convex spaces, since the construction is simply a straight line.

Definition 1.1.1. Let (X, x) be a pointed space. Then the *first fundamental group of X based at x* is the set $\mathcal{T}op_*(S^1, (X, x))/\sim$ of loops based at x (the chosen basepoint for S^1 is irrelevant), modulo basepoint-preserving homotopy. Group multiplication is given by concatenation of homotopy classes, i.e. $[f][g] = [fg]$, where the juxtaposition fg denotes concatenation.

We recall the pasting lemma for later use.

Lemma 1.1.2 (Pasting). *Let $A, B \subseteq X$ be both closed or both open subsets of a space, and let $X = A \cup B$. If $f : A \rightarrow Y$ and $g : B \rightarrow Y$ are maps such that $f|_{A \cap B} = g|_{A \cap B}$, then the continuous map $f \cup g : X \rightarrow Y$ defined in the usual way is continuous.*

Remark 1.1.3. It is not immediately clear why we require A, B be open or closed. Indeed, universality of pushout induces a map $f \cup g : A \amalg_{A \cap B} B \rightarrow Y$, and it is well known that $A \amalg_{A \cap B} B \cong$

$A \cup B$. However, this isomorphism only holds as *sets*, and since the forgetful functor $U : \mathcal{Top} \rightarrow \mathcal{Set}$ is *right* adjoint (not left), we don't get a nice statement about the topology on X for free. And this is where the issue lies: the topology on $A \amalg_{A \cap B} B$ is not necessarily the same as that on X (here A, B have the subspace topology), *unless* they are either both open or both closed. Given an open $U \subseteq A \amalg_{A \cap B} B$, we immediately have that $U \cap A$ and $U \cap B$ are open. If A, B are not open in X , then even if U is open in X , we may not have $U \cap A$ and $U \cap B$ open in X . Hence, the homeomorphism need not exist. It should be very clear why the closed/open hypotheses fix this issue.

I recall a few constructions of pointed spaces that may be useful later.

Definition 1.1.4. Let (X, x) and (Y, y) be pointed spaces. The coproduct in $\mathcal{Top}_*$, known as the *wedge sum*, is given by

$$X \vee Y := \frac{X \amalg Y}{x \sim y}.$$

Hence, we attach disjoint copies of X and Y at their basepoint.

Definition 1.1.5. Let (X, x) and (Y, y) be pointed spaces, and consider the product $X \times Y$. Note that $X \times \{y\} \cong X$ and $\{x\} \times Y \cong Y$ intersect only at the point (x, y) , meaning we have an embedding $X \vee Y \subseteq X \times Y$. Define the *smash product*

$$X \wedge Y := \frac{X \times Y}{X \vee Y}.$$

This is the product in $\mathcal{Top}_*$.

Lecture 2: The first fundamental group

Tue 08 Sep 2026 09:31

The prof rambled about the fundamental group. I knew all of it already, so I spent my time working through more useful stuff (that I also already knew).

As should be expected of a 2-cell, there are both vertical composites and horizontal composites of homotopies. We will now construct both.

Construction 1.1.6 (Vertical composition of homotopies). Let $f, g, h : X \rightarrow Y$ be maps, and let $H : f \Rightarrow g$, $K : g \Rightarrow h$ be homotopies. Define the homotopy $K \circ H : f \Rightarrow h$ by

$$(K \circ H)(x, t) = \begin{cases} H(x, 2t) & 0 \leq t \leq 1/2 \\ K(x, 2t - 1) & 1/2 \leq t \leq 1 \end{cases}.$$

This is well-defined if $K(-, 2t - 1)$ and $H(-, 2t)$ agree at $t = 1/2$. Indeed, we have

$$H(-, 2/2) = H(-, 1) = g,$$

$$K(-, 1 - 2/2) = K(-, 0) = g.$$

Finally, note that

$$(K \circ H)(-, 0) = H(-, 0) = f, \quad (K \circ H)(-, 1) = K(-, 2 - 1) = K(-, 1) = h.$$

Thus, $K \circ H$ is indeed a homotopy $h \simeq f$.

Construction 1.1.7 (Horizontal composition of homotopies). Let $f_0, f_1 : X \rightarrow Y$ and $g_0, g_1 : Y \rightarrow Z$ be continuous maps of spaces. Let $F : f_0 \Rightarrow f_1$ and $G : g_0 \Rightarrow g_1$ be homotopies. We will construct a homotopy $GF = G \cdot F : g_0 \circ f_0 \Rightarrow g_1 \circ f_1$, as depicted by the diagram

$$\left(X \begin{array}{c} \xrightarrow{f_0} \\ \Downarrow F \\ \xrightarrow{f_1} \end{array} Y \begin{array}{c} \xrightarrow{g_0} \\ \Downarrow G \\ \xrightarrow{g_1} \end{array} Z \right) = \left(X \begin{array}{c} \xrightarrow{g_0 \circ f_0} \\ \Downarrow GF \\ \xrightarrow{g_1 \circ f_1} \end{array} Z \right).$$

To define GF , let $\Delta : I \rightarrow I \times I$ be the diagonal map $t \mapsto (t, t)$, and let $\alpha : X \times (I \times I) \rightarrow (X \times I) \times I$ denote the associator $(x, (s, t)) \mapsto ((x, s), t)$. Then we define GF as the composite

$$X \times I \xrightarrow{1_X \times \Delta} X \times (I \times I) \xrightarrow{\alpha} (X \times I) \times I \xrightarrow{F \times 1_I} Y \times I \xrightarrow{G} Z.$$

On elements, this maps

$$(x, t) \xrightarrow{1_X \times \Delta} (x, (t, t)) \xrightarrow{\alpha} ((x, t), t) \xrightarrow{F \times 1_I} (F(x, t), t) \xrightarrow{G} G(F(x, t), t).$$

It's trivial to see that $1, \Delta, \alpha$ are continuous, and F, G are continuous by assumption. Thus, GF is continuous. It suffices to show it is a homotopy as claimed. Indeed,

$$GF(x, 0) = G(F(x, 0), 0) = G(f_0(x), 0) = g_0(f_0(x)) \implies GF(-, 0) = g_0 \circ f_0,$$

$$GF(x, 1) = G(F(x, 1), 1) = G(f_1(x), 1) = g_1(f_1(x)) \implies GF(-, 1) = g_1 \circ f_1.$$

Thus, we have horizontal composites.

Note that horizontal composition is associative, since it is essentially function composition. However, vertical composition is non-associative since it requires a reparameterization. Thus, $\mathcal{Top}$ is not a 2-category. The following proposition shows that it is associative up to homotopy.

Proposition 1.1.8. *Let $f : I \rightarrow X$ be a path, and let $\varphi : I \rightarrow I$ be a map sending $0 \mapsto 0$ and $1 \mapsto 1$. Then there is a homotopy to the reparameterization*

$$f \circ \varphi \simeq f.$$

Proof. We define

$$\Phi(t, s) = (1 - t)\varphi(s) + ts$$

which is a linear homotopy $\varphi \simeq 1$. We then horizontally compose the homotopies

$$I \begin{array}{c} \xrightarrow{\varphi} \\ \Downarrow \Phi \\ \xrightarrow{1} \end{array} I \begin{array}{c} \xrightarrow{f} \\ \Downarrow 1 \\ \xrightarrow{f} \end{array} X$$

to yield a homotopy $f \simeq f \circ \varphi$. ■

This motivates the fact that $\mathcal{Top}$ can be viewed as an ∞ -category. Usually, we identify along the Quillen equivalence $\mathcal{Top} \simeq \mathbf{sSet}$, and note that $\mathcal{Top}$ maps into the fibrant objects $\mathbf{sSet}^f \cong \mathcal{Kan}$. We then define the ∞ -category of spaces $\mathcal{S} := \mathcal{N}(\mathcal{Kan})$, for $\mathcal{N}$ the homotopy coherent nerve of the $\mathbf{sSet}$ -category $\mathcal{Kan} \subseteq \mathbf{sSet}$.

Ok now back to the lecture.

Proposition 1.1.9. *Let X be a path-connected space. Then $\pi_1(X, x_0)$ is invariant under choice of x_0 .*

Proof. Given $x, y \in X$ basepoints, by path-connectedness of X , there is a path $f : x \rightarrow y$ in X . This map induces the isomorphism on $\pi_1(X, x) \cong \pi_1(X, y)$ via mapping a loop $[\varphi] \in \pi_1(X, x)$ to the loop $[f \cdot \varphi \cdot f^{-1}]$, where f^{-1} denotes running backwards along f , and $\cdot$ is loop concatenation. That this is an isomorphism is obvious. ■

Definition 1.1.10. Let X be a space. Then X is *simply connected* if X is path connected, and $\pi_1(X) \cong 0$.

Discussion 1.1.11. We can translate all of this discussion into sSet. Recall that $|\Delta^2| \cong S^1$, and more generally $|\Delta^{n+1}| \cong S^n$. Given a simplicial set $S_\bullet$, define the first fundamental groupoid $\Pi_1(S)$ (warning: see following paragraph) to take S_0 as objects, and to take simplicial homotopy classes of maps $f : \Delta^1 \rightarrow S$ as morphisms in the following way. Let $\sigma = f_1([1])$ be the 1-simplex in S that fully determines S . Then if $d_0(\sigma) = y$ and $d_1(\sigma) = x$ are the vertices of σ , then we write $[\sigma] : x \rightarrow y$ in $\Pi_1(S)$ for $[\sigma]$ the homotopy class (with fixed endpoints) of f . Note here that $\Pi_1(S) = \mathbf{h}S$, for $\mathbf{h}$ the homotopy category. Given a vertex $s \in S_0$, define $\pi_1(S, s) := \Pi_1(s, s)$. Note that this can be identified with homotopy classes of maps $f : \partial\Delta^2 \rightarrow S$ based at s in the sense that $f_0([0]) = s$.

This turns out to be poorly behaved in general. In particular, $\Pi_1(S)$ need not be a category! Since morphisms in Π_1 are essentially morphisms in a quasi-category, we want to define a composite of $\sigma : x \rightarrow y$ and $\tau : y \rightarrow z$ as a filler for the inner horn $(\tau, \bullet, \sigma)$, which is precisely the datum of a 2-simplex $\Sigma \in S_2$ whose faces yield the diagram, together with the dotted arrow

$$\begin{array}{ccc} & y & \\ \sigma \nearrow & & \searrow \tau \\ x & \cdots \cdots \cdots & z \end{array},$$

rendering the diagram commutative. Of course, this need not exist. If S is a quasi-category, then such a filler exists. This promotes $\Pi_1(S)$ to a category (in particular, the homotopy category $\mathbf{h}S$ of the ∞ -category S), but not quite a groupoid.

We also need the datum of an inverse for all maps. In particular, given the horn indicated by the solid arrows

$$\begin{array}{ccc} & y & \\ \sigma \nearrow & & \searrow \sigma^{-1} \\ x & \xrightarrow{1} & x, \end{array}$$

we would require the existence of a dotted arrow σ^{-1} , providing an inverse to σ . Thus, we need a filler for $(\bullet, 1, \sigma)$. We similarly want fillers for $(\sigma, 1, \bullet)$. Thus, S should be a Kan complex, promoting S to an ∞ -groupoid, and thus $\Pi_1(S) = \mathbf{h}S$ to a groupoid. This provides the necessary group structure for $\pi_1(S)$. The higher homotopy groups are defined similarly, using maps $\partial\Delta^{n+1} \rightarrow S$.

Lecture 3: Fundamental group of S^1

Thu 10 Sep 2026 09:33

Definition 1.1.12. Let X be a space. A *covering space* of X is a space $\tilde{X}$, together with a continuous map $p : \tilde{X} \rightarrow X$, such that for all $x \in X$, there is an open neighborhood $x \in U \subseteq X$ such that

$p^{-1}(U)$ can be realized as a disjoint union

$$p^{-1}(U) = \coprod_{\alpha \in J} \tilde{U}_\alpha,$$

for $\tilde{U}_\alpha \subseteq \tilde{X}$ open subsets with the property that the restriction $p|_{\tilde{U}_\alpha} : \tilde{U}_\alpha \cong U$ is a homeomorphism.

Intuition 1.1.13. The best intuition is the canonical picture of $\mathbb{R}$ twisted into a coil, sitting above S^1 . Choose a point, and a small neighborhood. Then at each level of the coil, you can lift it to a homeomorphic open. Thus you get a disjoint union of opens, all homeomorphic to U .

Lemma 1.1.14. *Let $p : \tilde{X} \rightarrow X$ be a covering space. Then any path $f : I \rightarrow X$ starting at $x_0 \in X$, and any $\tilde{x}_0 \in p^{-1}(x_0)$, there is a unique lift $\tilde{f} : I \rightarrow \tilde{X}$. More generally, given a homotopy H of paths in X which preserves endpoints, we can lift H to a homotopy $\tilde{H}$ of the lifted paths in $\tilde{X}$.*

Theorem 1.1.15. *For any $p \in S^1$, we have $\pi_1(S^1, p) \cong \mathbb{Z}$.*

Lecture 4

Tue 22 Sep 2026 09:31

I missed 2 classes. Apparently we are now up to Van Kampen's theorem.

Theorem 1.1.16 (Van Kampen). *Let X be the union of path-connected opens A_α each containing the basepoint $x \in X$, such that each $A_\alpha \cap A_\beta$ is path-connected. Then the map from the free product $\Phi : *_\alpha A_\alpha \rightarrow X$ induced by the inclusion maps is surjective. If each triple intersection $A_\alpha \cap A_\beta \cap A_\gamma$ is also path-connected, Then the kernel is generated by $i_{\alpha\beta}(\omega)i_{\beta\alpha}(\omega)^{-1}$ for $\omega \in \pi_1(A_\alpha \cap A_\beta)$, and $i_{\alpha\beta} : \pi_1(A_\alpha \cap A_\beta) \hookrightarrow \pi_1(A_\alpha)$ induced by the inclusion.*

Construction 1.1.17. A CW complex is similar to the realization of a simplicial set. Start with an $\mathbb{N}$ -graded set $X^\bullet = \{X^n\}_{n \geq 0}$. We think of an element $e^n \in X_n$ as an open n -ball B^n , and call it an n -cell.

Start by realizing X^0 as a discrete set of points. We write the realization of X^0 as $|X^0|$. For every 1-cell $e_\alpha^1 \in X^1$, we require a map $\varphi_\alpha^1 : \partial B^1 \rightarrow |X^0|$. We then construct $|X^1|$, the realization of all X^n with $n \leq 1$, by gluing the 1-cells in along these maps:

$$|X^1| = \left(|X^0| \amalg \coprod_{\alpha} B_{\alpha}^1 \right) / \langle \forall x \in \partial B_{\alpha}^1, x \sim \varphi_{\alpha}^1(x) \rangle.$$

We continue this process inductively:

$$|X^n| = \left(|X^{n-1}| \amalg \coprod_{\alpha} B_{\alpha}^n \right) / \langle \forall x \in \partial B_{\alpha}^n, x \sim \varphi_{\alpha}^n(x) \rangle.$$

If $X^\bullet$ has only finitely many nonempty sets, then this is sufficient, as the realization $X = |X^\bullet|$ is just $|X^n|$ for some n . However, if it is infinite, then we note that there are inclusions $|X^n| \subseteq |X^{n+1}|$ for all n , and define X as the colimit over these inclusions.

We typically denote $|X^n|$ by X^n , and call it the n -skeleton. I only separated X^n and $|X^n|$ for simplicity.

Note that CW complexes are Hausdorff, locally contractible, and locally path connected.

Proposition 1.1.18. *Let X be a space.*

1. *If Y is obtained by attaching 2-cells to X , then the inclusion $X \subseteq Y$ induces a surjection $\pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$ whose kernel is generated by elements $\gamma_\alpha \varphi_\alpha \gamma_\alpha^{-1}$, for γ some path $x_0 \rightarrow s$ with $s \in S^2$ a basepoint, and φ_α the attaching map.*
2. *If Y is obtained by attaching n -cells to X for $n > 2$, then $X \subseteq Y$ induces an isomorphism $\pi_1(X, x_0) \cong \pi_1(Y, y_0)$.*
3. *If X is a path-connected CW complex, then the inclusion of the 2-skeleton induces an isomorphism $\pi_1(X^2, x_0) \cong \pi_1(X, x_0)$.*

1.2 Covering spaces

Definition 1.2.1. Recall definition 1.1.12. Given an object X , we define a category of covering spaces of X . Given covering spaces A, B of X , a map $A \rightarrow B$ is defined as a continuous map which commutes with the covering maps. In other words, a map φ such that the diagram

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & B \\ p_A \searrow & & \swarrow p_B \\ & X & \end{array}$$

commutes. The isomorphisms are precisely when φ is a homeomorphism. Automorphisms are also referred to as *deck transformations*. We do not often require a notation for this category, but we may refer to it as $\text{Cover}(X)$.

Remark 1.2.2. With definition 1.1.12 as Hatcher gave it, the covering map p need not be surjective. Let $x \in X$, and let $U \subseteq X$ be open. Then the preimage could be empty. By taking the index set J to be empty, we have vacuously that for all $\alpha \in J$, the restriction $p|_{\tilde{U}_\alpha} \cong U$ is a homeomorphism. This is valid, since we require

$$p^{-1}(U) = \coprod_{\alpha \in J} \tilde{U}_\alpha,$$

and the empty disjoint union is the colimit over 0 which is the initial object, hence $\emptyset$ in $\mathbf{Top}$.